MRINALI CHARHATE

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PROFESSIONAL SUMMARY

Computer Science undergraduate with experience in cybersecurity, machine learning, NLP, and generative AI.

EDUCATION

Vellore Institute of Technology, Vellore 2023 - 2027
BTech in Computer Science and Engineering (Information Security)
CGPA: 9.15/10.0

SKILLS

Programming:	Python, C, C++, SQL
AI/ML:	TensorFlow, PyTorch, Hugging Face, LangChain, Pandas, NumPy
Security Tools:	Nessus, Burp Suite, Nmap, Wireshark
Cloud & Backend:	AWS (Lambda, ECS, DynamoDB), FastAPI, Docker, REST APIs, Git
Languages Known:	English, Hindi, Marathi

EXPERIENCE

Cybersecurity Intern May 2026 - July 2026
Abhita Aerospace Pvt. Ltd. *Mumbai, Maharashtra*

- Identified 35+ vulnerabilities across 5 applications using Nessus, prioritizing findings by severity.
- Conducted penetration testing (Burp Suite) and network analysis (Nmap, Wireshark), uncovering 12 risks.
- Reviewed security logs, flagging 28+ anomalies that directly informed remediation actions.

AI-ML Intern June 2025
Reliance New Energy *Mumbai, Maharashtra*

- Collaborated with 3 teams to define requirements, aligning pipelines and retrieval workflows with business goals.
- Built hybrid retrieval models (semantic + keyword), improving query accuracy from 78% to 91%.
- Developed dashboards visualizing energy trends and retrieval metrics, enabling faster stakeholder decisions.

PROJECTS

SentinelAI *SentinelAI*
Python, FastAPI, React, TypeScript, LangChain, Elasticsearch, Ollama, Docker

- Built AI-powered SOC platform integrating Splunk, Elasticsearch, and Wazuh, detecting 15+ event types.
- Implemented LLM-assisted incident summarization with MITRE ATT&CK mapping, cutting triage time by 60%.
- Developed 30+ reusable React components across six security operations modules powering analyst workflows.

Video Frame Reconstruction *Video Reconstruction*
Python, OpenCV, NumPy, NetworkX, Multiprocessing

- Reconstructed shuffled video via graph optimization and MST, achieving 95%+ frame-order accuracy.
- Optimized sequencing with OpenCV and multiprocessing, reducing reconstruction time by 45%.
- Processed 300-frame videos end-to-end in under 30 seconds using multi-threaded similarity computation.

SecureSense *SecureSense*
Python, DistilBERT, Hugging Face, FastAPI, PyTorch, Docker

- Developed hybrid PII detection (DistilBERT, regex, heuristics), achieving 96% precision across 8 categories.
- Built FastAPI inference service with overlap-safe masking, processing documents in under 300ms.
- Containerized the service with Docker, enabling consistent deployment across environments.

PATENT AND CERTIFICATIONS

Real-Time Multimodal System for Adaptive Emotion-Aware Hindi-English Communication (Published Patent) *Application No.: 202641033897*
Machine Learning Specialization (Stanford/DeepLearning.AI)
Supervised Machine Learning | *Advanced Learning Algorithms*
IBM Gen-AI *IBM GEN-AI Certificate*